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Shortest Path Finding

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Project phase 2

1. Introduction

Graph theory is a fundamental area of study in computer science and mathematics, focusing on the relationships and connections between objects represented as vertices and edges. One of the key problems in graph theory is the **Shortest Path Finding** problem, which seeks to identify the most efficient route between two nodes in a graph. This problem has significant implications in various applications, including transportation networks, telecommunications, and social network analysis. Understanding how to effectively navigate through graphs to find the shortest path is crucial for optimizing resources and improving performance in these domains.

2. Shortest Path Finding in Unweighted Graphs

The problem of **Shortest Path Finding** involves determining the most efficient route between two specified nodes (or vertices) within an unweighted graph. An unweighted graph is defined as a graph in which all edges have the same weight or cost, typically assumed to be one. This characteristic simplifies the traversal process as it eliminates the need for cost-based comparisons between paths.

To solve this problem, graph traversal algorithms such as **Breadth-First Search (BFS)** are employed. BFS is particularly well-suited for unweighted graphs due to its systematic exploration of nodes level by level, ensuring that the first time a node is reached, it is via the shortest possible path.

During the traversal, it is crucial to maintain a record of each node's predecessor or parent. This allows for the reconstruction of the shortest path once the destination node is reached. By backtracking from the destination to the start node using the recorded parent information, the complete path can be efficiently reconstructed.

While **Depth First Search (DFS)** can also be used to find a path between two nodes in an unweighted graph, it is generally less efficient for this specific problem. DFS explores as far down a branch as possible before backtracking, which may lead to longer paths being discovered first. Although it can find a path, it does not guarantee that this path is the shortest. To use DFS effectively for shortest path finding, one would need to explore all potential paths and keep track of the shortest one encountered, which can result in higher computational costs and longer execution times compared to BFS. Thus, while DFS is a viable option, it is typically not preferred for shortest path finding in unweighted graphs.

The significance of this problem lies not only in its theoretical implications within graph theory but also in its practical applications across various domains, including navigation systems, network routing, and resource allocation problems. Understanding and effectively implementing algorithms for shortest path finding is essential for optimizing performance in these applications.

3. Theoretical Implementation

3.1 Breadth-First Search (BFS) :

a. How BFS Works for Shortest Path

BFS is a highly effective algorithm for determining the shortest path in unweighted graphs. It systematically explores neighboring nodes in a level-order fashion, ensuring that when a node is encountered for the first time, it is reached via the shortest possible route.

The Breadth-First Search algorithm adheres to the following essential steps for implementing the shortest path search in unweighted graphs:

1. **Initialization :** Start by initializing a queue to keep track of nodes that need to be explored. Additionally, maintain a dictionary to record the predecessor of each node that is visited.
2. **Traversal:** Begin the traversal from the starting node by adding it to the queue and marking it as visited. As long as the queue is not empty, repeatedly dequeue a node and explore its unvisited neighbors.
3. **Path Reconstruction:** If the target node is reached during the traversal, backtrack using the predecessor information to reconstruct the shortest path.
4. **Termination:** If the queue is exhausted without reaching the target node, it indicates that there is no path between the starting and target nodes.

b. BFS Pseudocode

```
function bfs_shortest_path(graph, start, target):
    initialize queue with start node
    initialize visited set
    initialize parent dictionary to reconstruct the path

    while queue is not empty:
        current_node = dequeue from queue
        if current_node is target:
            return reconstruct_path(parent, start, target)
        if current_node is not in visited:
            add current_node to visited
            for each neighbor in graph[current_node]:
                if neighbor is not in visited:
                    enqueue neighbor to queue
                    set parent[neighbor] = current_node
return "No path found"
```

c. Time Complexity and Space Complexity :

Time Complexity: O(V + E), where V is the number of vertices and E is the number of edges. Each vertex and edge is processed once.

Space Complexity: O(V), for the visited set and the queue.

3.2 Depth-First Search (DFS)

a. How DFS Works for Shortest Path

DFS can effectively identify paths between nodes; however, it does not provide a guarantee for the shortest path. This algorithm explores as deeply as possible along a branch before backtracking.

The Depth-First Search algorithm can be utilized to find paths in graphs, although it does not ensure that these paths are the shortest. The implementation involves the following key steps::

1. Initialization: Start by initializing a stack (or using recursion) to keep track of nodes to be explored. A dictionary is used to record the predecessor of each visited node.

2. Traversal: Begin from the starting node, pushing it onto the stack and marking it as visited. Continue this process by popping a node from the stack and exploring its unvisited neighbors.

3. Path Reconstruction: If the target node is reached, backtrack using the predecessor information to reconstruct the path.

4. Termination: The algorithm continues until all possible paths are explored or the target node is found. If the stack is exhausted without finding the target, it indicates that no path exists.

b. DFS Pseudocode

```
function dfs_shortest_path(graph, current_node, target, visited, path):
    add current_node to visited
    append current_node to path

    if current_node is target:
        return path
    for each neighbor in graph[current_node]:
        if neighbor is not in visited:
            result = dfs_shortest_path(graph, neighbor, target, visited, path)
            if result is not None:
                return result

    path.pop()  // backtrack
    return None
```

c. Time Complexity and Space Complexity :

Time Complexity: O(V + E) in the worst case, as it may visit every vertex and edge.
Space Complexity: O(V) for the visited set and the recursion stack.

4. Code Implementation

The implementation of the Shortest Path Finding problem leverages graph traversal algorithms such as Breadth-First Search (BFS) and Depth-First Search (DFS). The provided code defines two functions: bfs_shortest_path and dfs_shortest_path. Both algorithms operate on unweighted graphs, meaning that all edges are considered equal in weight, allowing them to efficiently find paths between vertices based solely on connectivity.

The BFS algorithm explores nodes level by level using a queue. It systematically visits each vertex's neighbors, ensuring that once the destination node is reached, the shortest path is reconstructed by backtracking through a recorded set of predecessor nodes. This process guarantees that the first time a node is encountered, it represents the shortest path to that node.

In contrast, the DFS algorithm explores as far as possible along each branch before backtracking, utilizing a stack for its traversal. While it can find a path to the destination, it does not guarantee the shortest path because it may explore longer routes first before reaching the goal.

Both algorithms are tested on two different types of unweighted graphs: one generated with random integers and the other with random character vertices. The measure_performance function assesses the execution time and memory usage of each algorithm, providing insights into their efficiency. This comprehensive approach highlights the differences in performance characteristics between BFS and DFS for the shortest path finding problem.

4.1 Source code

To access source code: see the code/ folder of this repository.

4.2 Implementation of Random Character Graph and Shortest Path Algorithms

This code generates a random undirected graph where vertices are represented by characters (A, B, C, ..., Z, AA, AB, etc.) and finds the shortest paths between them using Breadth-First Search (BFS) and Depth-First Search (DFS). The graph is structured as an adjacency list, created by randomly selecting pairs of vertices while avoiding self-loops and duplicate edges. The BFS algorithm explores vertices level by level using a queue, tracking visited nodes to reconstruct the shortest path, with a time complexity of O(V + E). In contrast, the DFS algorithm uses a stack to delve deeply into paths before backtracking, also with a time complexity of O(V + E), but it does not guarantee the shortest path. The program measures the performance of both algorithms in terms of execution time and memory usage and generates a random graph with 100 vertices and 300 edges for testing, showcasing the practical applications of graph theory in computer science.

Function Name	Description	Time Complexity
generate_random_graph(num_vertices, num_edges)	Generates a random undirected graph with a specified number of vertices and edges.	O(E) Complexity is driven by edge generation to ensure uniqueness, with potential multiple iterations in the worst case.
draw_graph(graph)	Prints the adjacency list representation of the graph.	O(V) Iterates through all vertices to print connections, resulting in linear complexity relative to the number of vertices.
bfs_shortest_path(graph, start, end)	Performs BFS to find the shortest path between two vertices in the graph.	O(V + E) Explores each vertex and edge once, with queue operations being O(1), leading to linear complexity.
dfs_shortest_path(graph, start, end)	Performs DFS to find a path (not necessarily the shortest) between two vertices in the graph.	O(V + E) Like BFS, it processes each vertex and edge once, with stack operations being O(1).
measure_performance(search_algorithm, graph, start, end)	Measures the execution time and memory usage of the specified search algorithm.	O(V + E) Dominated by the search algorithm (BFS or DFS), with negligible additional time for execution measurement.
calculate_vertices_and_edges(graph)	Calculates the total number of vertices and edges in the graph.	O(V) Counts vertices and edges by iterating through the adjacency list, resulting in linear complexity.

Table 1 Function Descriptions and Time Complexity Analysis

4.3 Implementation of Random Numbers Graph and Shortest Path Algorithms

This code generates a random undirected graph represented as an adjacency list, where vertices are indexed by integers. It finds the shortest paths between vertices using Breadth-First Search (BFS) and Depth-First Search (DFS). The generate_random_graph function creates edges while avoiding self-loops and duplicates. BFS explores vertices level by level using a queue, tracking visited nodes for path reconstruction, with a time complexity of O(V + E). DFS, on the other hand, uses a stack to explore paths deeply before backtracking, also with a time complexity of O(V + E), but it does not ensure the shortest path. The program measures execution time and memory usage for both algorithms and generates a test graph with 100 vertices and 200 edges, demonstrating graph theory applications in computer science.

Function Name	Description	Time Complexity
generate_random_graph(num_vertices, num_edges)	Generates a random undirected graph with a specified number of vertices and edges. Returns an adjacency list representation of the graph.	O(E) Complexity arises from generating unique edges, with time varying based on the number of edges relative to vertices.
bfs_shortest_path(graph, start, end)	Performs a Breadth-First Search (BFS) to find the shortest path between two nodes in an unweighted graph.	O(V + E) Every vertex and edge is visited at most once, resulting in linear complexity.
dfs_shortest_path(graph, start, end)	Performs a Depth-First Search (DFS) to find a path between two nodes in an unweighted graph.	O(V + E) Similar to BFS, DFS explores each vertex and edge, leading to linear time complexity.
measure_performance(search_algorithm, graph, start, end)	Measures the time taken and memory usage of a given search algorithm (BFS or DFS) on the graph.	O(V + E) Complexity is determined by the search algorithm (BFS or DFS) called within the function.
calculate_vertices_and_edges(graph)	Calculates the number of vertices (V) and edges (E) in the graph.	O(V) Iterates through the adjacency list to count vertices and edges, visiting each vertex once.

Table 2 Function Descriptions and Time Complexity Analysis

4.4 Comparison of Algorithms

Based on outputs from code , we can draw several comparisons between the BFS and DFS algorithms

a. Implementation of Random Character Graph and Shortest Path Algorithms

Example One

random graph with 100 vertices and 300 edges

```
BG: CN, C, BT, BX
C: BG, T, DC, CK, DI, BO, DT
BD: CM, J
CM: BD, BE, BH, BI, BN, CD, S
DK: BW, DQ, Z, CK, O
BI: CZ, U, K, BA, CM, CH, DM, CC
BN: CK, DB, V, CM, BQ
BP: CU, U, BM, X
CU: BP, BM, S, BT, CQ, DR
DN: DA, T, DC, CF
CJ: DA, I, V
Z: DK, P, CE, BC
CP: BH, CG, CN, BY
CE: Z, BR, DR, G
BF: CB, CL, DT, B
BQ: O, BN, G, CH, N, K

=======  BFS and DFS Algorithms  =======

        Start Node: H, End Node: R

 -----------------

BFS Shortest path from H to R:

        Path: ['H', 'T', 'R']
        Time taken: 0.000073 seconds
        Memory usage: 7602 bytes
        Time Complexity: O(V + E)
        Time Complexity: O(100 + 300)

 -----------------

DFS Shortest path from H to R:

        Path: ['H', 'B', 'BF', 'DT', 'C', 'BO', 'L', 'DR', 'CU', 'BT', 'R']
        Time taken: 0.000132 seconds
        Memory usage: 7602 bytes
        Time Complexity: O(V + E)
        Time Complexity: O(100 + 300)
```

Table 3 Random Character Graph Example One

Example Two

random graph with 100 vertices and 300 edges

```
BG: CO, Y, BD, CT, CP
CO: BG, P, C, Y, U, CC, DU
P: CO, T, CM, BP, DM, CZ
CF: CU, CH, DE, DN, CB, CQ
BO: BT, BB
BT: BO, BF, CT, BN, CD, DJ
CT: BC, BX, DD, BN, BT, N, Y, BG
DG: DF, BA, DK, DJ, BS, CE, DU, U
DI: CJ, Z, BB, CL, CU, BU
CP: C, DQ, BF, BQ, F, CK, CH, BG
CD: CU, CW, CC, CS, BT
CW: CD, BB, CE
BP: CB, DQ, P
DQ: CP, BP, DD
BU: DD, Z, DI
CV: DO, Q, R
CH: CF, Z, O, N, BY, CP
F: BX, DB, CP, CA, BK
CQ: DD, CS, CF
DU: CI, DH, A, BA, DG, CO
V: BS, DB
BK: CB, BY, F
BJ: Z, BD
DT: BW, N, DJ
CN: X, DJ, DA
BR: H

=======  BFS and DFS Algorithms  =======

        Start Node: BV, End Node: DH

 -----------------

BFS Shortest path from BV to DH:

        Path: ['BV', 'U', 'T', 'DH']
        Time taken: 0.000250 seconds
        Memory usage: 7602 bytes
        Time Complexity: O(V + E)
        Time Complexity: O(100 + 300)

 -----------------

DFS Shortest path from BV to DH:

        Path: ['BV', 'M', 'DE', 'BN', 'I', 'DC', 'CB', 'DO', 'A', 'CL', 'CM', 'BB', 'CK', 'CP', 'CH', 'N', 'CI', 'DM', 'T', 'DH']
        Time taken: 0.000041 seconds
        Memory usage: 7602 bytes
        Time Complexity: O(V + E)
        Time Complexity: O(100 + 300)
```

Table 4 Random Character Graph Example Two

The key differences between the two examples are as follows: In the first example, BFS found a shorter path ['H', 'T', 'R'] in 0.000073 seconds, while DFS found a longer path in 0.000132 seconds. In the second example, BFS found a path ['BV', 'U', 'T', 'DH'] in 0.000250 seconds, while DFS found a longer path in 0.000041 seconds, making it faster. The memory usage and time complexity remained the same for both algorithms in both examples. The efficiency of the algorithms was reversed, with BFS being faster in the first example and DFS being faster in the second example.

Criteria	BFS (Example One)	DFS (Example One)	BFS (Example Two)	DFS (Example Two)
Path Found	['H', 'T', 'R']	['H', 'B', 'BF', 'DT', 'C', 'BO', 'L', 'DR', 'CU', 'BT', 'R']	['BV', 'U', 'T', 'DH']	['BV', 'M', 'DE', 'BN', 'I', 'DC', 'CB', 'DO', 'A', 'CL', 'CM', 'BB', 'CK', 'CP', 'CH', 'N', 'CI', 'DM', 'T', 'DH']
Time Taken	0.000073 seconds	0.000132 seconds	0.000250 seconds	0.000041 seconds
Memory Usage	7602 bytes	7602 bytes	7602 bytes	7602 bytes
Time Complexity	O(V + E)	O(V + E)	O(V + E)	O(V + E)
Time Complexity	O(100 + 300)	O(100 + 300)	O(100 + 300)	O(100 + 300)
Efficiency	Faster in Example One	Slower in Example One	Slightly slower than DFS	Faster in Example Two
Shortest Path	Yes	No	Yes	Yes

Table 5 BFS vs. DFS Performance Summary - Random Character Graph -

The comparative analysis of the Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms highlights key differences in performance across two examples. In the first example, BFS found the shorter path ['H', 'T', 'R'] in 0.000073 seconds, while DFS took 0.000132 seconds for a longer path. Conversely, in the second example, DFS located a longer path in 0.000041 seconds, whereas BFS required 0.000250 seconds to find the path ['BV', 'U', 'T', 'DH']. Notably, memory usage and time complexity remained consistent for both algorithms, adhering to the O(V+E) relationship, suggesting that performance differences stemmed from the graph's specific characteristics rather than resource utilization. This underscores the importance of understanding the problem context and selecting the appropriate algorithm based on specific requirements.

In the context of the random character graph, the results demonstrate the trade-offs between the BFS and DFS algorithms in terms of path length, execution time, and consistency. While both algorithms were able to find the shortest path in the second example, the first example illustrates how BFS can outperform DFS in identifying the optimal solution, even though DFS may be more efficient in certain scenarios. This underscores the need for a nuanced approach to algorithm selection and the value of empirical analysis in determining the most suitable approach for a given problem.

b. Implementation of Random Numbers Graph and Shortest Path Algorithms

Example One

random graph with 100 vertices and 200 edges

```
72: [48, 80, 32]
73: [18, 32, 78]
74: [36, 79]
75: [93, 91, 76]
76: [91, 75, 58, 62]
77: [29, 35, 7, 43]
78: [63, 49, 73, 35]
79: [69, 74]
80: [47, 54, 66, 72, 17, 14, 34, 29]
81: [8, 84, 61, 65]
82: [39, 93]
83: [36]
84: [28, 99, 81, 90, 70, 6, 31, 45]
85: [38, 93]
86: [28, 19, 51, 34, 65]
87: [49, 44, 59, 40, 61]
88: [11, 20, 36, 38, 65, 3]
89: [35, 42, 38]
90: [70, 84, 21, 66]
91: [76, 75, 95, 94, 13, 23, 44, 92, 67, 33]
92: [59, 12, 91, 53, 69]
93: [75, 82, 64, 66, 52, 44, 85]
94: [21, 91, 48, 33]
95: [61, 91]
96: [9]
97: [62, 13, 38, 52]
98: [13, 41, 44, 68]
99: [65, 0, 84, 8]

======== BFS and DFS Algorithms ========


 -----------------
BFS Shortest path from 38 to 43:

        Path: [38, 26, 71, 43]
        Time taken: 0.000152 seconds
        Memory usage: 7488 bytes
        Time Complexity: O (V + E)
        Time Complexity: O(100 + 200)


 -----------------
DFS Shortest path from 38 to 43:

        Path: [38, 32, 23, 42, 56, 49, 78, 35, 77, 43]
        Time taken: 0.000051 seconds
        Memory usage: 7488 bytes
        Time Complexity: O (V + E)
        Time Complexity: O(100 + 200)
```

Table 6 Random Number Graph Example One

Example Two

random graph with 100 vertices and 200 edges

```
72: [69, 54]
73: [95, 9, 80]
74: [22, 7, 31, 99]
75: [33, 7, 8]
76: [61]
77: [87, 83, 50]
78: [30, 20, 7]
79: [98, 51, 84, 8, 10]
80: [73, 31, 8]
81: [71, 48, 4, 47, 30]
82: [12, 51, 3, 50]
83: [27, 43, 47, 77, 31, 85]
84: [44, 97, 66, 54, 79, 59, 56]
85: [19, 26, 58, 15, 98, 83]
86: [95, 55]
87: [55, 77, 59]
88: [20, 49, 47, 43, 59]
89: [63, 32, 70]
90: [91, 15, 2]
91: [90, 9]
92: [95, 54, 42, 96]
93: [70, 61, 22, 30, 94, 32]
94: [2, 0, 5, 14, 93, 21, 34, 3]
95: [86, 18, 92, 73]
96: [12, 50, 49, 36, 17, 92]
97: [84, 60, 71, 4]
98: [3, 63, 79, 43, 59, 85]
99: [26, 32, 29, 12, 25, 74]

======== BFS and DFS Algorithms ========


 -----------------
BFS Shortest path from 37 to 67:

        Path: [37, 32, 4, 64, 67]
        Time taken: 0.000036 seconds
        Memory usage: 7488 bytes
        Time Complexity: O (V + E)
        Time Complexity: O(100 + 200)


 -----------------
DFS Shortest path from 37 to 67:

        Path: [37, 9, 49, 69, 54, 47, 81, 30, 43, 11, 28, 67]
        Time taken: 0.000075 seconds
        Memory usage: 7488 bytes
        Time Complexity: O (V + E)
        Time Complexity: O(100 + 200)
```

Table 7 Random Number Graph Example Two

The comparative analysis of the BFS and DFS algorithms in the numerical graph reveals key differences in their performance. In the first example, BFS found the shorter path faster, while in the second example, DFS located a longer path more efficiently. Despite consistent memory usage and time complexity, the performance differences stemmed from the graph's specific characteristics rather than resource utilization. This underscores the importance of understanding the problem context and selecting the appropriate algorithm based on the specific requirements, emphasizing the need for a nuanced approach to algorithm choice.

Criteria	BFS (Example One)	DFS (Example One)	BFS (Example Two)	DFS (Example Two)
Path Found	[37, 32, 6, 67]	[37, 9, 49, 69, 54, 47, 81, 30, 43, 11, 28, 67]	[38, 26, 71, 43]	[38, 32, 2005, 42, 56, 49, 78, 35, 77, 43]
Time Taken	0.00036	0.000075	0.000152	0.000051
Memory Usage	7488 bytes	7488 bytes	7488 bytes	7488 bytes
Time Complexity	O(V + E)	O(V + E)	O(V + E)	O(V + E)
Time Complexity	O(100 + 200)	O(100 + 200)	O(100 + 200)	O(100 + 200)
Efficiency	Faster in Example One	Slower in Example One	Slightly slower than DFS	Faster in Example Two
Shortest Path	Yes	No	Yes	No

Table 8 BFS vs. DFS Performance Summary - Random Number Graph -

In the context of the numerical graph, the results demonstrate the trade-offs between the BFS and DFS algorithms in terms of path length, execution time, and consistency. In the first example, the BFS algorithm found the shorter path [37, 32, 4, 64, 67] more efficiently, taking only 0.000036 seconds, while the DFS algorithm took longer at 0.000075 seconds to find a more extended path [37, 9, 49, 69, 54, 47, 81, 30, 43, 11, 28, 67]. However, in the second example, the DFS algorithm was able to locate a longer path [38, 32, 23, 42, 56, 49, 78, 35, 77, 43] in a shorter time of 0.000051 seconds, compared to the 0.000152 seconds required by the BFS algorithm to find the shorter path [38, 26, 71, 43].

The comparative analysis of the Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms in the random number graph reveals distinct performance trade-offs. In the first example, BFS was able to find the shorter path faster than DFS, which took longer to locate a more extended path. However, the roles were reversed in the second example, where DFS efficiently found a longer path in less time compared to BFS.

5. Conclusion

The comparative analysis of Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms across randomly generated number and character graphs provides valuable insights into their nuanced performance in shortest path finding. Overall, the comparison illustrates the nuances of each algorithm's performance. BFS generally excels in finding the shortest path in unweighted graphs, as evidenced by its faster execution times in specific instances. However, DFS can be faster in certain scenarios, particularly when it reaches the target node without needing to explore all possible paths.

The results highlighted the trade-offs between BFS and DFS; while BFS was able to find the shorter path more efficiently in some instances, DFS outperformed BFS in others by locating longer paths in shorter times. Across both graph types, consistent memory usage and time complexity of O(V + E) for both algorithms suggest that performance differences are primarily driven by the specific graph characteristics rather than resource utilization. This underscores the importance of understanding the specific characteristics of the graph when selecting the appropriate algorithm.

BFS consistently demonstrates superior performance in identifying the shortest path due to its level-order traversal, making it particularly suitable for applications such as network routing and navigation systems. However, the choice of algorithm should not be based solely on general performance characteristics; it must be guided by the specific requirements of the problem. While BFS is optimal for finding the shortest path, it may not be the best choice in scenarios where memory usage is a concern, as it can consume more memory due to storing all nodes at the current level. Conversely, DFS, with its depth-first traversal, can be more memory-efficient and advantageous for problems requiring exhaustive exploration, such as puzzle solving or pathfinding in less structured environments.

Empirical insights gained from testing algorithms on varied graph structures are invaluable for making informed decisions. Factors such as graph density, size, and connectivity can significantly influence performance, leading to scenarios where DFS outperforms BFS, particularly in sparse graphs or when the goal is a valid path rather than the shortest one. Ultimately, the effectiveness of BFS or DFS hinges on aligning the choice of algorithm with the specific needs of the application. By leveraging empirical insights from randomized experiments, developers and researchers can better understand the trade-offs associated with each algorithm, enabling them to select the most appropriate method for their use case. This nuanced approach not only enhances performance but also contributes to the overall efficiency and effectiveness of graph traversal solutions across various fields, from artificial intelligence to transportation systems.

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